Zero-Logit Model Provenance Auditing in Mission-Critical Financial AI: Privacy-Preserving Black-Box Attribution via Prober-Generator Resonance Spectroscopy

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ABSTRACT

In mission-critical financial AI workflows—spanning synthetic fraud generation, anti-money laundering (AML) report synthesis, automated loan underwriting, and customer-facing banking agents—institutions increasingly enforce strict **inference privacy** by withholding token-level log-probabilities (logits) and internal model states to prevent data exfiltration, model distillation, and membership inference. However, logit suppression cripples conventional white-box forensic auditing tools (e.g., DetectGPT, Binoculars), leaving compliance auditors reliant on brittle surface stylometrics or uncalibrated scalar thresholds. In this paper, we introduce **Prober-Generator Resonance Spectroscopy**, a privacy-preserving, zero-logit black-box audit framework designed to verify financial generative AI provenance purely through text-only multi-sentence cloze infilling ($k \in \{2, 3, 4\}$) and permutation-invariant Kuhn-Munkres bipartite matching in $O(k^3)$ time. We demonstrate why fixed binary thresholds ($\tau = 28.0\%$) suffer domain calibration drift in specialized financial and regulatory prose, and replace thresholding with continuous resonance margins ($\Delta\mu = \mu_{\text{AI}} - \mu_{\text{Human}}$) and threshold-independent separability (AUROC/AUPRC). Evaluated across an $N = 600$ four-slice benchmark including high-stakes decision reasoning and ethical deliberation, our framework achieves **80.68% pairwise AUROC** on DeepSeek/Qwen ($\mu = 45.65\% \pm 17.76\%$), **77.00% pairwise AUROC** on Meta LLaMA-3.3 ($\mu = 49.30\% \pm 25.42\%$), **75.76% AUROC** (**94.91% AUPRC**) in ethical conundrums, and **75.69% pooled AUROC** (**92.99% AUPRC**) across four canonical instruction-tuned families ($N = 344$), establishing an auditable, non-invasive compliance mechanism for real-world banking infrastructure.

KEYWORDS

Financial AI Security, Zero-Logit Auditing, Model Provenance, Inference Privacy, Black-Box Forensics, Cloze Infilling, Bipartite Matching.

1 INTRODUCTION: FINANCIAL AI PRIVACY VS. AUDITABILITY

The deployment of Large Language Models (LLMs) across mission-critical financial services—including anti-money laundering (AML) suspicious activity report (SAR) generation, automated loan decisioning, customer support, and algorithmic trading advisories—presents a sharp trade-off between **inference privacy** and **forensic provenance auditability**. Exposing token-level log-probabilities (logits) at financial API boundaries enables adversaries to perform logit-inversion attacks, proprietary model extraction, and training-data membership inference against sensitive transaction histories [1]. Consequently, financial institutions, payment gateways, and banking APIs strictly withhold logits, returning only raw output text strings.

However, existing zero-shot AI detectors such as DetectGPT [2], Fast-DetectGPT, and Binoculars require white-box access to token log-probabilities or cross-model perplexity ratios. Under black-box API constraints, risk auditors and compliance officers are forced to rely on surface stylometric heuristics (which fail on formal financial prose [**?**]) or uncalibrated scalar cutoffs.

1.1 Why Fixed Scalar Thresholds Drift in Financial Domains

In multi-scale cloze infilling, dense cosine similarity C_{dense} is coupled with novel relational predicate overlap P_{novel} via a sigmoidal dynamic gate ($w_{\text{base}} = 0.20, w_{\text{boost}} = 0.40, \kappa = 15.0, c_0 = 0.70$):

$$w_{\text{dense}}(C) = w_{\text{base}} + \frac{w_{\text{boost}} - w_{\text{base}}}{1 + \exp\left(-\kappa(C - c_0)\right)}. \tag{1}$$

When an uncalibrated inflection center ($c_0 = 0.70$) and a static cutoff ($\tau = 28.0\%$) are applied across heterogeneous financial domains—such as contractual legalities or standardized banking procedures—domain-specific terminology shifts human representation similarity toward c_0. This induces domain-dependent calibration drift, confounding authentic model provenance with technical vocabulary density. We address this by formalizing **Prober-Generator Resonance Spectroscopy**, evaluating continuous resonance distributions ($\mu \pm \sigma$), inter-class separation margins ($\Delta\mu = \mu_{\text{AI}} - \mu_{\text{Human}}$), and threshold-independent separability (AUROC, AUPRC) across an $N = 600$ forensic benchmark.

2 METHODOLOGY: ZERO-LOGIT PURE CLOZE RESONANCE SPECTROSCOPY

Let $D = (s_1, s_2, \ldots, s_L)$ denote a financial document or transaction narrative partitioned into $L \geq 6$ contiguous sentences. Our framework enforces a strict **Zero-Logit, Zero-Internal-State Invariant**: no token probabilities, hidden activations, or lexical frequency counters are accessed.

2.1 Multi-Scale Contiguous Void Infilling ($k \in \{2, 3, 4\}$)

To isolate macro-discursive propositional structure from local token phrasing, we excise contiguous multi-sentence voids of length $k \in \{2, 3, 4\}$:

$$B_{t,k} = (s_t, \ldots, s_{t+k-1}), \quad D_{\setminus(t,k)} = (s_1, \ldots, s_{t-1}, \texttt{[VOID_}k\texttt{]}, s_{t+k}, \ldots, s_L). \tag{2}$$

A black-box reference prober P_{prober} (`nvidia/nemotron-3-super-120b`, temperature $T = 0.0$) receives only the text context $D_{\setminus(t,k)}$ and generates a k-sentence infill hypothesis $\hat{B}_{t,k} = (\hat{s}_1, \ldots, \hat{s}_k)$.

2.2 Permutation-Invariant Bipartite Matching & Cross-Span Minimum Gate

Because an infilling prober may generate valid financial reasoning steps in a permuted clause sequence $\pi \in \mathcal{S}_k$, strict index-by-index comparison underestimates structural congruence. We construct a $k \times k$ pairwise similarity matrix $W_{ij} = \Phi(s_{t+i-1}, \hat{s}_j)$ combining dense semantic embeddings with novel relational predicate overlap P_{novel} (excluding surrounding context tokens in $D_{\setminus(t,k)}$), and compute the optimal assignment via the Kuhn-Munkres (Hungarian) algorithm in $O(k^3)$ time (< 0.1 ms):

$$C_k(B_{t,k}, \hat{B}_{t,k}) = \max_{\pi \in S_k} \frac{1}{k} \sum_{i=1}^{k} \Phi\left(s_{t+i-1}, \hat{s}_{\pi(i)}\right). \tag{3}$$

Across evaluated document zones $z \in \{1, \ldots, Z\}$ and void horizons $k \in \{2, 3, 4\}$, let p_k denote scale-specific resonance and $\bar{C} = \frac{1}{3}(p_2 + p_3 + p_4)$ the multi-scale cloze mean. To capture human cognitive variance—where even a single idiosyncratic rhetorical pivot collapses the prober's predictability—we integrate the **Cross-Span Minimum Gate** $S_{\min} = \min_{z,k} C_k^{(z)}$ into the continuous V3 Resonance score $\mathcal{R}_{\text{V3}}(D) \in [0, 100]$.

3 EMPIRICAL EVALUATION ON $N = 600$ FOUR-SLICE BENCHMARK

We evaluate our framework on an $N = 600$ forensic benchmark comprising four balanced slices of $n = 150$ documents each (512 AI texts across 6 foundation model families and 88 human controls, 22 per slice).

Table 1: Continuous Resonance Spectroscopy and Threshold-Independent Separability Across Thematic Slices ($N = 600$; 512 AI, 88 Human Controls).

Thematic Slice	Total (n)	AI / Hu	AI ($\mu \pm \sigma$)	Human ($\mu \pm \sigma$)	AUROC	AUPRC
Slice 1: Abstract Foundations	150	128 / 22	36.37% ± 22.7%	24.90% ± 16.9%	64.58%	92.12%
Slice 2: Ethical Conundrums	150	128 / 22	**42.93%** ± **22.9%**	**21.70%** ± **17.3%**	**75.76%**	**94.91%**
Slice 3: Strategic Decisions	150	128 / 22	40.69% ± 22.6%	24.31% ± 19.3%	**70.67%**	**93.48%**
Slice 4: Day-to-Day Applied	150	128 / 22	39.32% ± 22.9%	28.93% ± 15.4%	62.93%	91.99%
Combined $N = 600$ Benchmark	**600**	**512 / 88**	**39.83%** ± **22.9%**	**24.96%** ± **17.5%**	**68.76%**	**93.21%**

3.1 Key Findings in Mission-Critical Contexts

- **High Separability on Critical Deliberations:** In Slice 2 (Ethical Conundrums, Table 1), AI models follow structured reasoning templates resulting in high resonance ($\mu_{\text{AI}} = 42.93\%$), whereas human deliberation exhibits sharp ideological divergences ($\mu_{\text{Human}} = 21.70\%$), producing +21.23% separation and **75.76% AUROC** (**94.91% AUPRC**).
- **Robust Pairwise Separability across Major AI Providers:** As shown in Table 2, DeepSeek/Qwen achieves **80.68% AUROC** and Meta LLaMA-3.3 achieves **77.00% AUROC**. Across the Top-4 instruction-tuned families ($N = 344$), pooled separability reaches **75.69% AUROC** (**92.99% AUPRC**).
- **Absence of Prober Bias:** Although the prober is `nemotron-3-super-120b`, the NVIDIA cohort scores lowest (19.13%), while Meta (49.30%) and DeepSeek/Qwen (45.65%) score highest, proving that resonance measures the generator's discursive determinism rather than superficial architectural kinship.

4 OPERATIONAL DEPLOYMENT & LIMITATIONS IN FINANCIAL SYSTEMS

For operational banking deployment: (1) querying a 120B reference prober across three multi-sentence void horizons requires 2.4–4.0 s per passage, designating this method as an **Asynchronous Tier-2 Audit Pipeline** (e.g., automated AML batch verification, loan documentation auditing) rather than a real-time gateway packet filter; (2) under strict zero-logit constraints, pooled 6-family separability is 68.76% (75.69% on Top-4 families), providing transparent, auditable bipartite matching matrices rather than opaque scalar cutoffs.

5 MATHEMATICAL INVARIANCE PROOF

Theorem 1 (Permutation-Invariant Bipartite Optimality). *Let $B = (s_1, \ldots, s_k)$ be an excised discourse span and $\hat{B} = (\hat{s}_1, \ldots, \hat{s}_k)$ be the prober infill. For any sentence permutation $\sigma \in \mathcal{S}_k$ producing $\hat{B}' = (\hat{s}_{\sigma(1)}, \ldots, \hat{s}_{\sigma(k)})$, the Kuhn-Munkres operator C_k satisfies:*

$$C_k(B, \hat{B}') = C_k(B, \hat{B}). \tag{4}$$

Proof. Because $\mathcal{S}_k$ is a finite symmetric group under functional composition, the mapping $\tau = \sigma \circ \pi$ is a bijection on $\mathcal{S}_k$. Hence $\max_{\pi \in \mathcal{S}_k} \frac{1}{k} \sum_{i=1}^{k} \Phi(s_i, \hat{s}_{\sigma(\pi(i))}) = \max_{\tau \in \mathcal{S}_k} \frac{1}{k} \sum_{i=1}^{k} \Phi(s_i, \hat{s}_{\tau(i)}) = C_k(B, \hat{B})$. By the Birkhoff-von Neumann Theorem, this is solved in $O(k^3)$ polynomial time. □

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Table 2: Signature Decomposition and Pairwise Separability Across Provenance Families on $N = 600$ Benchmark.

Provenance Family	N	Scale p_2	Scale p_3	Scale p_4	Cloze Mean ($\bar{C}$)	Min Span ($S_{\min}$)	Novel Pred (P_{novel})	Fused V3 ($\mu \pm \sigma$)	$\Delta\mu$ vs. Hu	AUROC vs. Hu
Meta (LLaMA 3.3 70B)	84	31.81%	33.13%	33.41%	32.78%	21.91%	22.97%	**49.30% ± 25.42%**	**+24.34%**	77.00%
DeepSeek / Qwen (V3 / Qwen 2.5)	84	**41.81%**	**44.88%**	**46.41%**	**44.37%**	**27.91%**	**35.41%**	**45.65% ± 17.76%**	+20.69%	**80.68%**
Google (Gemini 2.5 Flash Lite)	88	31.24%	32.34%	33.33%	32.31%	21.27%	21.60%	**44.35% ± 24.06%**	+19.39%	72.97%
OpenAI (GPT-4o / 4o-mini)	88	33.49%	34.96%	35.66%	34.70%	22.62%	25.72%	**43.07% ± 23.62%**	+18.11%	72.41%
Anthropic (Claude 3.5 Sonnet)	84	31.68%	32.57%	33.25%	32.50%	22.97%	21.67%	**37.11% ± 18.86%**	+12.15%	68.11%
Top-4 Canonical Families Pooled	**344**	**34.48%**	**36.22%**	**37.10%**	**35.93%**	**23.39%**	**26.36%**	**45.56% ± 23.04%**	**+20.60%**	**75.69%**
Human Controls (Academic / Formal)	88	31.23%	31.90%	32.76%	31.96%	20.91%	20.49%	**24.96% ± 17.47%**	0.00%	Baseline Floor
NVIDIA (Nemotron 3 Cohort)	84	25.54%	24.90%	25.40%	25.28%	14.69%	13.73%	19.13% ± 9.56%	−5.83%	41.01%